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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("/content/global_food_waste.csv")
df
```

	Country	Year	Food Category	Total Waste (Thousand Tonnes)	Economic Loss (Million USD)	Population (Million)
0	France	2021	Prepared Foods	13254.57	47662.70	837.61
1	Pakistan	2024	Dairy Products	25666.20	22687.68	204.30
2	USA	2023	Prepared Foods	48645.94	50616.79	301.21
3	India	2021	Grains & Cereals	28614.04	28188.92	411.26
4	Australia	2021	Meat & Seafood	21486.28	29539.92	1100.32
...	...	...	...	...	...	...
95	Italy	2021	Grains & Cereals	47768.52	53854.77	640.64
96	Japan	2019	Meat & Seafood	34108.18	25743.03	325.01
97	Germany	2018	Meat & Seafood	49379.48	43097.06	752.85
98	India	2022	Meat & Seafood	32125.18	19173.63	190.79
99	USA	2022	Frozen Foods	47741.00	44640.29	860.71

100 rows × 6 columns

Next steps: Generate code with df New interactive sheet

```
df.shape
```

```
(100, 6)
```

```
df.head()
```

	Country	Year	Food Category	Total Waste (Thousand Tonnes)	Economic Loss (Million USD)	Population (Million)
0	France	2021	Prepared Foods	13254.57	47662.70	837.61
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4	Australia	2021	Meat & Seafood	21486.28	29539.92	1100.32

Next steps: Generate code with df New interactive sheet

```
df.columns
```

```
Index(['Country', 'Year', 'Food Category', 'Total Waste (Thousand Tonnes)',
       'Economic Loss (Million USD)', 'Population (Million)'],
      dtype='object')
```

Start coding or generate with AI.

COUNTRY-LEVEL INSIGHTS

```
country_waste = df.groupby("Country")["Total Waste (Thousand Tonnes)"].sum()
df
```

	Country	Year	Food Category	Total Waste (Thousand Tonnes)	Economic Loss (Million USD)	Population (Million)
0	France	2021	Prepared Foods	13254.57	47662.70	837.61
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...	...	...	...	...	...	...
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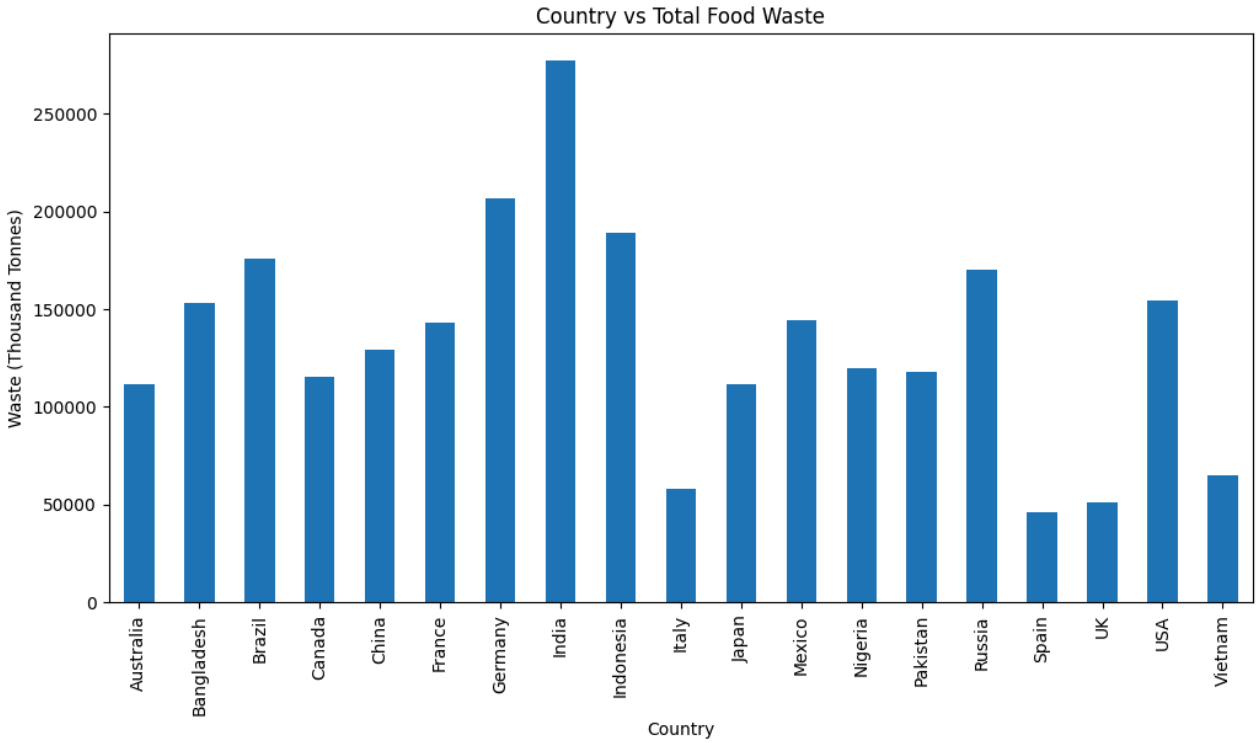
99	USA	2022	Frozen Foods	47741.00	44640.29	860.71

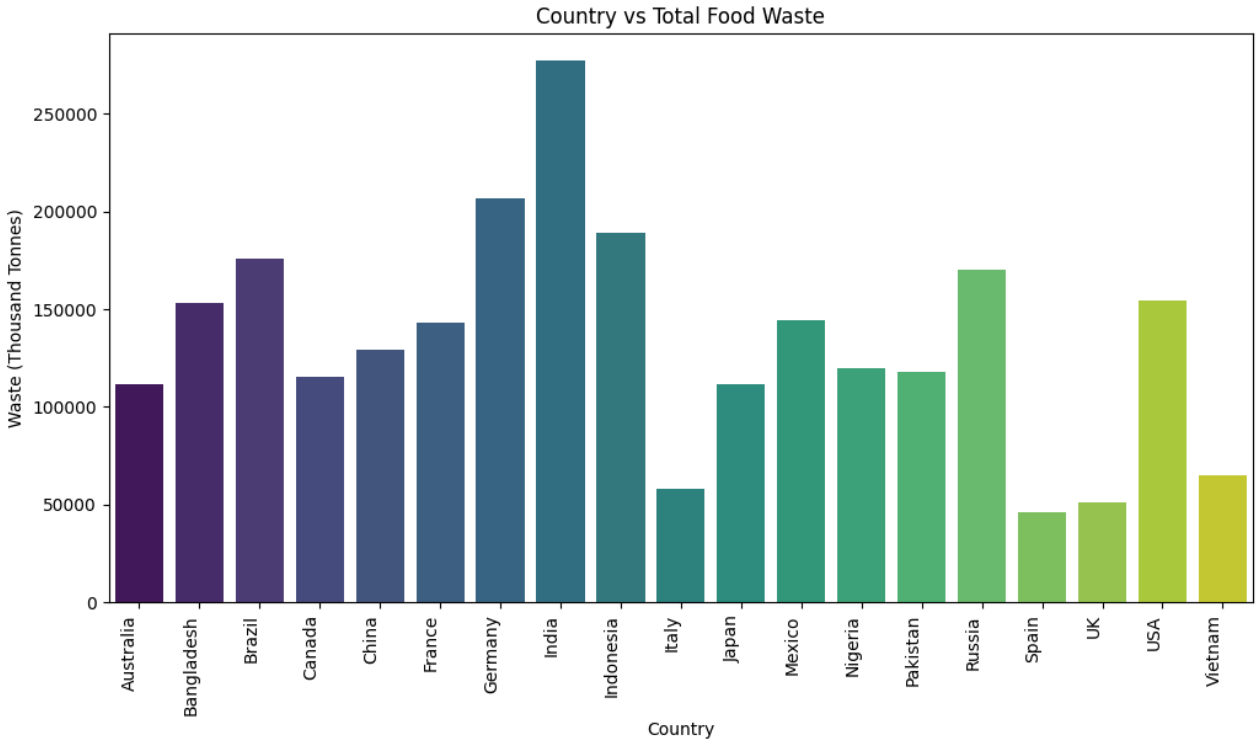
100 rows × 6 columns

Next steps: Generate code with df New interactive sheet

```
country_waste = df.groupby("Country")["Total Waste (Thousand Tonnes)"].sum()
plt.figure(figsize=(12, 6))
country_waste.plot(kind="bar")
plt.title("Country vs Total Food Waste")
plt.xlabel("Country")
plt.ylabel("Waste (Thousand Tonnes)")
plt.show()
```




```
plt.figure(figsize=(12, 6))
sns.barplot(x=country_waste.index, y=country_waste.values, hue=country_waste.index, palette='viridis')
plt.title("Country vs Total Food Waste")
plt.xlabel("Country")
plt.ylabel("Waste (Thousand Tonnes)")
plt.xticks(rotation=90, ha='right')
plt.show()
```




```
#sort by
country_waste_sorted = country_waste.sort_values(ascending=False)
display(country_waste_sorted.head())
```

Country	Total Waste (Thousand Tonnes)
India	277294.16
Germany	206825.81
Indonesia	188906.26
Brazil	175581.34
Russia	170229.14

dtype: float64

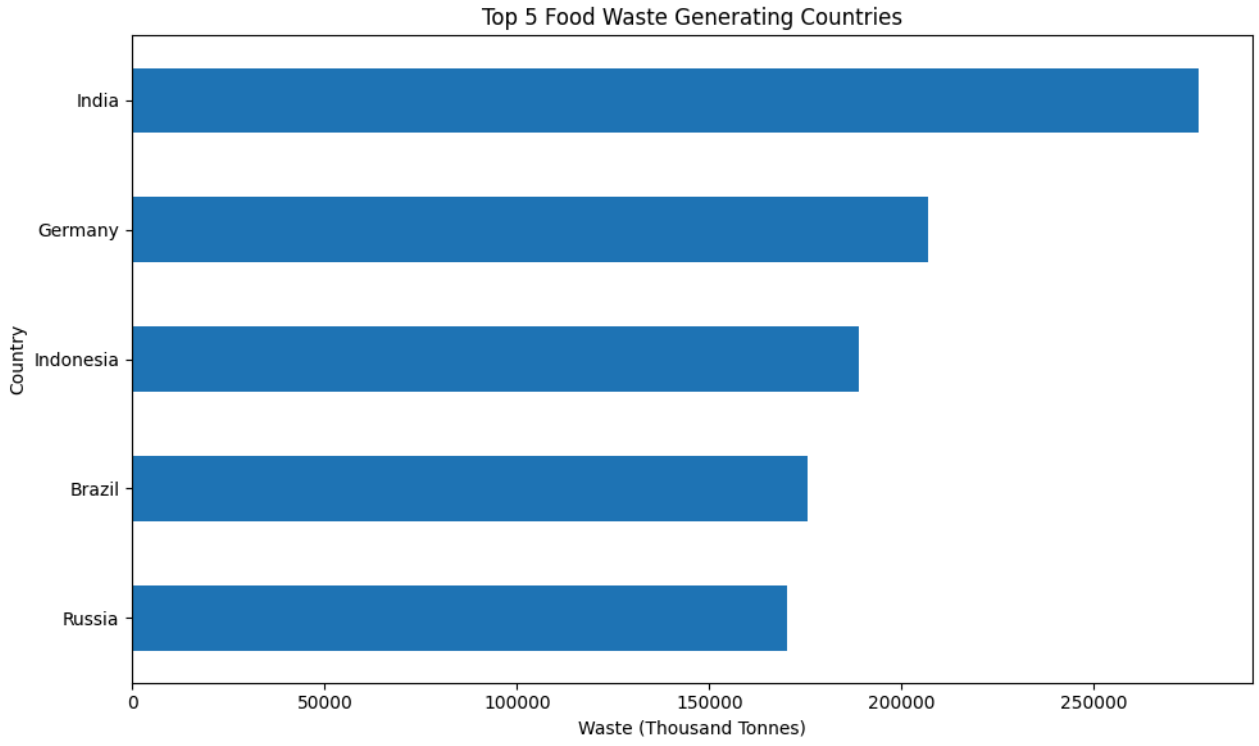
Which country generates maximum food waste?

```
#get top country
display(country_waste_sorted.head(1))
```

Country	Total Waste (Thousand Tonnes)
India	277294.16

dtype: float64

```
top5 = country_waste.sort_values(ascending=False).head(5)
plt.figure(figsize=(10, 6))
top5.plot(kind="barh")
plt.title("Top 5 Food Waste Generating Countries")
plt.xlabel("Waste (Thousand Tonnes)")
plt.ylabel("Country")
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```



Countries with high waste but low population

```
#Total waste per country
waste_by_country = df.groupby("Country")["Total Waste (Thousand Tonnes)"].sum()
display(waste_by_country.head())
```

Country	Total Waste (Thousand Tonnes)
Australia	111475.84
Bangladesh	152904.13
Brazil	175581.34
Canada	115145.03
China	129426.26

dtype: float64

```
#Average population per country
population_by_country = df.groupby("Country")["Population (Million)"].mean()
display(population_by_country.head())
```

Country	Population (Million)
Australia	568.992500
Bangladesh	857.407500
Brazil	1051.350000
Canada	852.380000
China	829.181667

dtype: float64

```
country_data = pd.concat([waste_by_country, population_by_country], axis=1)
display(country_data.head())
```

Country	Total Waste (Thousand Tonnes)	Population (Million)
Australia	111475.84	568.992500
Bangladesh	152904.13	857.407500
Brazil	175581.34	1051.350000
Canada	115145.03	852.380000
China	129426.26	829.181667

```
#sort by using values
country_data.sort_values("Total Waste (Thousand Tonnes)", ascending=False).head()
```

Country	Total Waste (Thousand Tonnes)	Population (Million)
India	277294.16	729.253333
Germany	206825.81	535.060000
Indonesia	188906.26	469.545714
Brazil	175581.34	1051.350000
Russia	170229.14	906.001667

Economic loss due to food waste

```
df["Economic_Loss"] = df["Total Waste (Thousand Tonnes)"] * 1000
loss_by_country = df.groupby("Country")["Economic_Loss"].sum()
loss_by_country.sort_values(ascending=False).head()
```

Country	Economic_Loss
India	277294160.0
Germany	206825810.0
Indonesia	188906260.0
Brazil	175581340.0
Russia	170229140.0

dtype: float64

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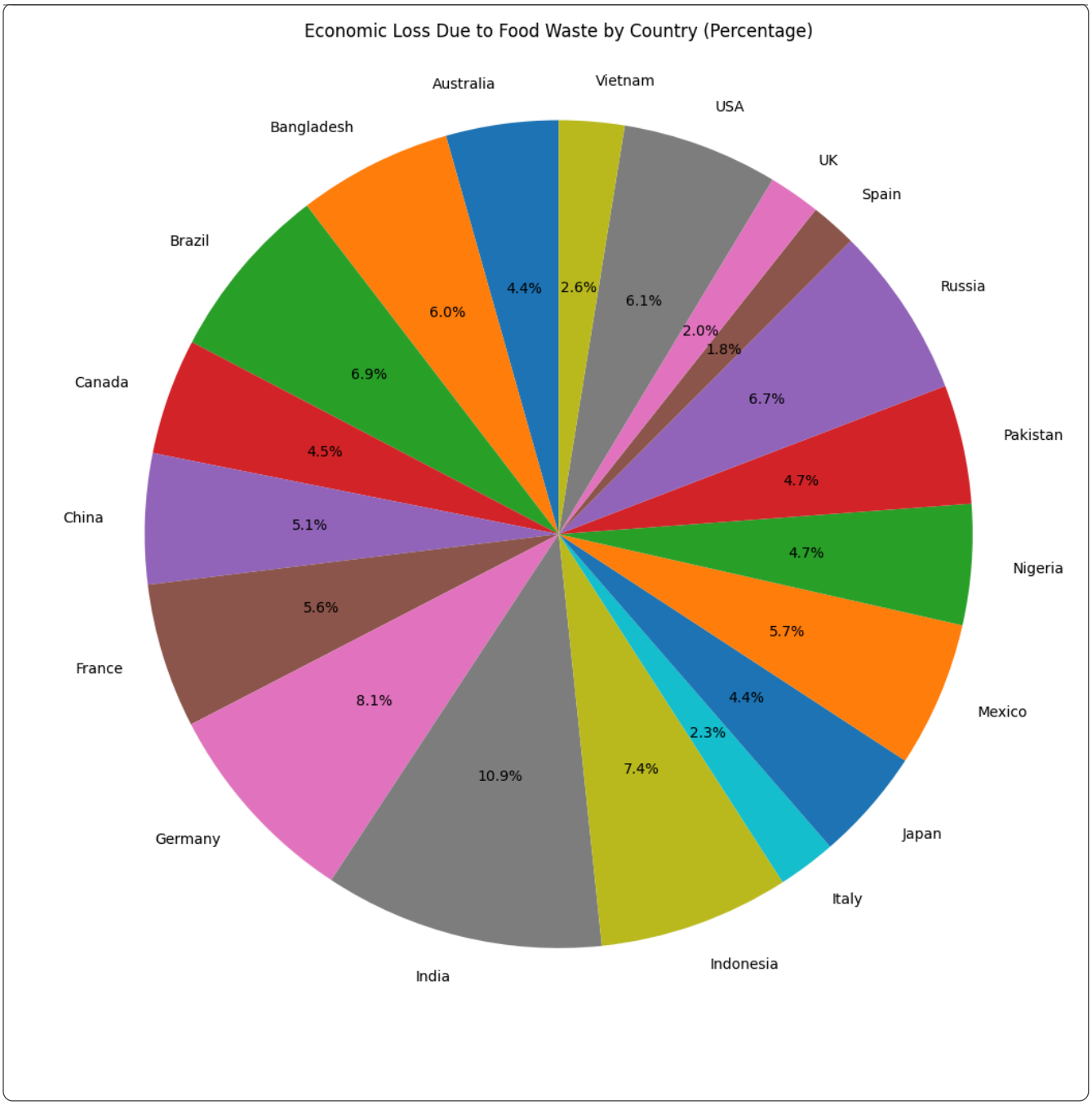
CATEGORY-LEVEL INSIGHTS

```
plt.figure(figsize=(10, 10))
plt.pie(loss_by_country, labels=loss_by_country.index, autopct='%1.1f%%', startangle=90)
plt.title("Economic Loss Due to Food Waste by Country (Percentage)")
plt.axis('equal')
plt.tight_layout()
plt.show()
```



Most wasted food categories globally !!

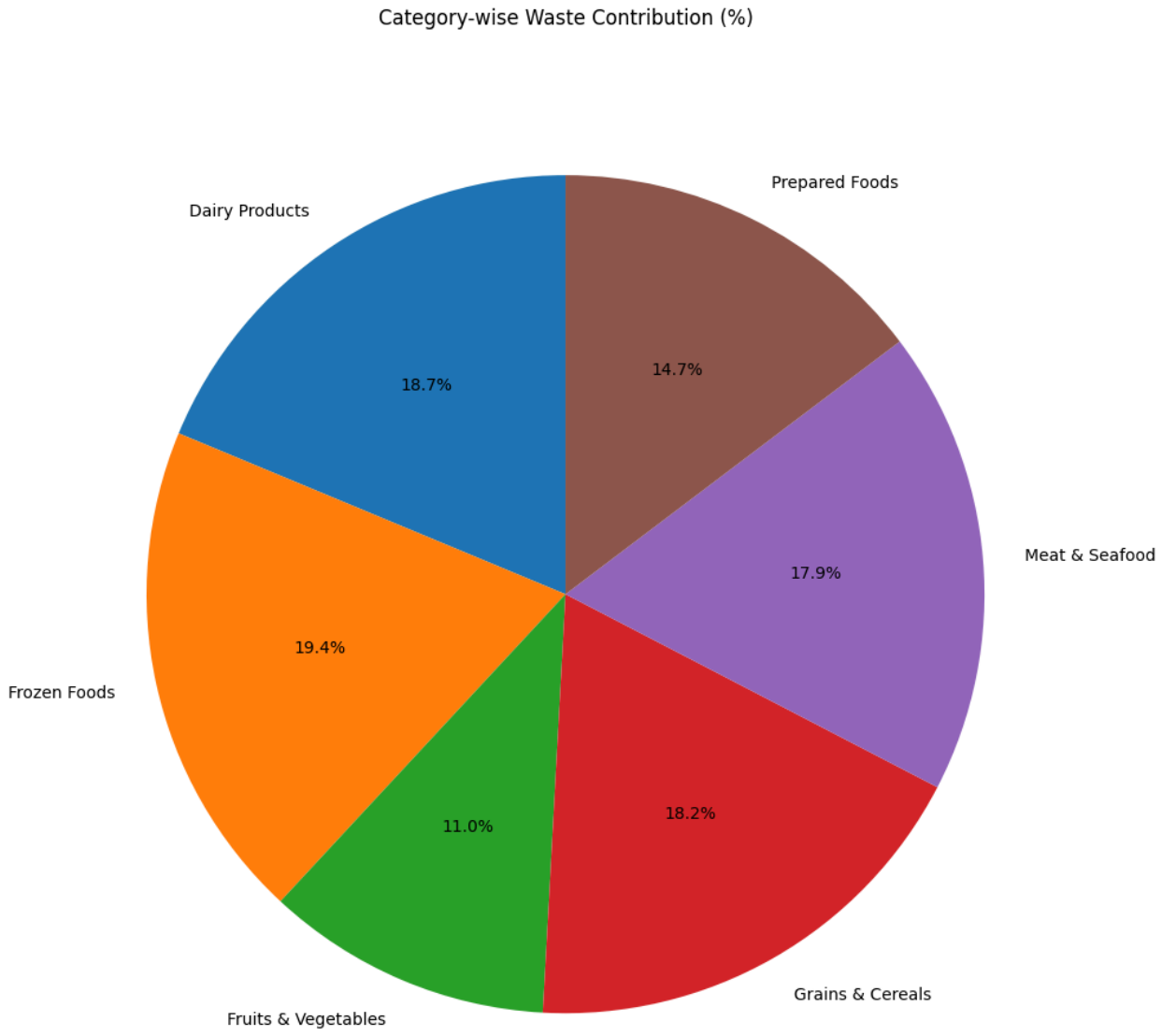
```
category_waste = df.groupby("Food Category")["Total Waste (Thousand Tonnes)"].sum()
display(category_waste.sort_values(ascending=False))
```

Food Category	Total Waste (Thousand Tonnes)
Frozen Foods	491770.45
Dairy Products	475589.54
Grains & Cereals	463339.06
Meat & Seafood	455153.56
Prepared Foods	373236.71
Fruits & Vegetables	280514.87

dtype: float64

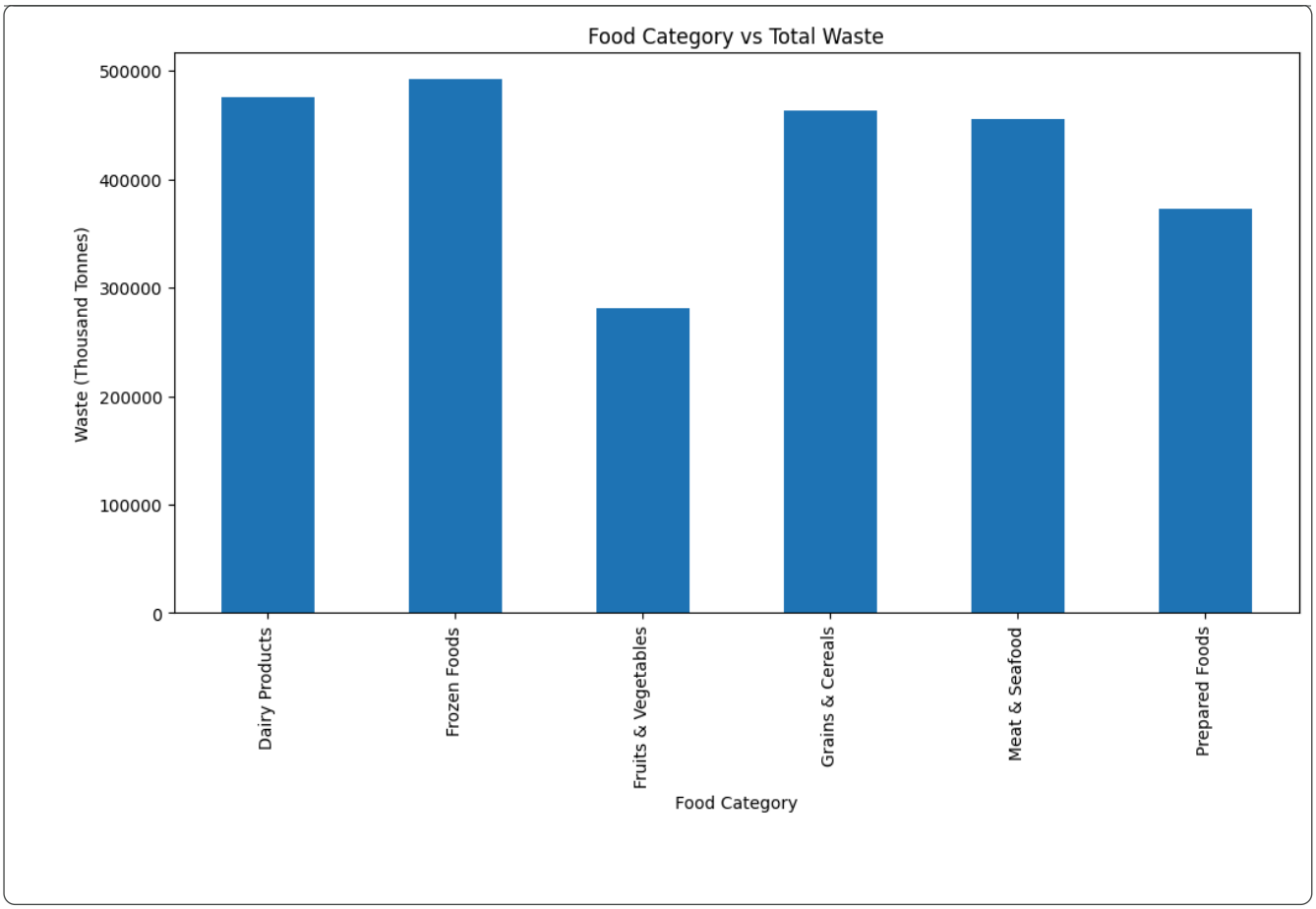
```
total_waste = df["Total Waste (Thousand Tonnes)"].sum()
category_percentage = (category_waste / total_waste) * 100
plt.figure(figsize=(10, 10))
plt.pie(category_percentage, labels=category_percentage.index, autopct="%1.1f%%", startangle=90)
plt.title("Category-wise Waste Contribution (%)")
plt.axis('equal')
```

```
plt.tight_layout()
plt.show()
```



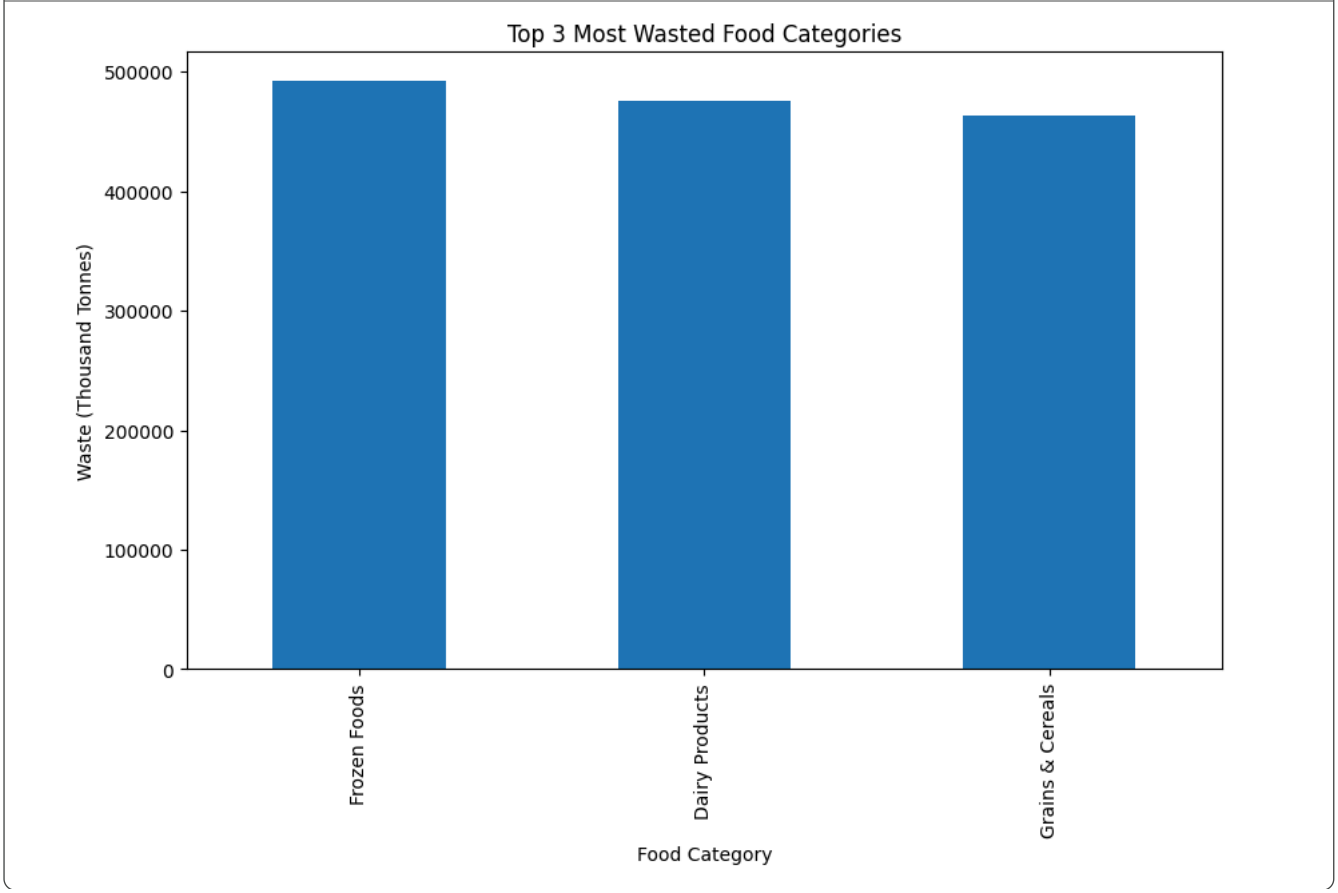

```
category_waste = df.groupby("Food Category")["Total Waste (Thousand Tonnes)"].sum()

plt.figure(figsize=(12, 6))
category_waste.plot(kind="bar")
plt.title("Food Category vs Total Waste")
plt.xlabel("Food Category")
plt.ylabel("Waste (Thousand Tonnes)")
plt.show()
```



Category-wise waste percentage contribution !!

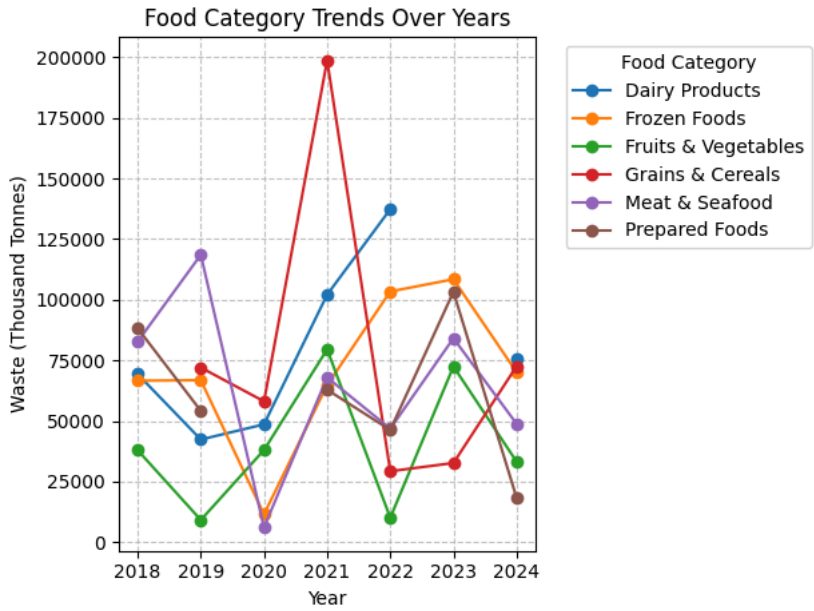
```
top3_categories = category_waste.sort_values(ascending=False).head(3)
plt.figure(figsize=(10, 6))
top3_categories.plot(kind="bar")
plt.title("Top 3 Most Wasted Food Categories")
plt.xlabel("Food Category")
plt.ylabel("Waste (Thousand Tonnes)")
plt.show()
```



```
category_year = df.groupby(["Year", "Food Category"])["Total Waste (Thousand Tonnes)"].sum()

plt.figure(figsize=(12, 7))
category_year.unstack().plot(marker='o')
plt.title("Food Category Trends Over Years")
plt.xlabel("Year")
plt.ylabel("Waste (Thousand Tonnes)")
plt.grid(True, linestyle='--', alpha=0.7)
plt.xticks(category_year.index.get_level_values('Year').unique())
plt.legend(title="Food Category", bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```

```
<Figure size 1200x700 with 0 Axes>
```




```
total_waste = df["Total Waste (Thousand Tonnes)"].sum()
category_percentage = (category_waste / total_waste) * 100
display(category_percentage.sort_values(ascending=False))
```

Food Category	Total Waste (Thousand Tonnes)
Frozen Foods	19.364059
Dairy Products	18.726916
Grains & Cereals	18.244538
Meat & Seafood	17.922224
Prepared Foods	14.696649
Fruits & Vegetables	11.045614

dtype: float64

Food Category vs Total Waste

```
category_waste = df.groupby("Food Category")["Total Waste (Thousand Tonnes)"].sum()
display(category_waste)
# This confirms the total waste for each food category, with Frozen Foods having the highest total.
```

Food Category	Total Waste (Thousand Tonnes)
Dairy Products	475589.54
Frozen Foods	491770.45
Fruits & Vegetables	280514.87
Grains & Cereals	463339.06
Meat & Seafood	455153.56
Prepared Foods	373236.71

dtype: float64

Start coding or generate with AI.

TIME-BASED INSIGHTS

Year-wise global food waste trend

```
yearly_waste = df.groupby("Year")["Total Waste (Thousand Tonnes)"].sum()
display(yearly_waste)
#It appears that 2021 had the highest amount of food waste, while 2020 had the lowest in this dataset.
```

Year	Total Waste (Thousand Tonnes)
2018	344893.05
2019	363129.46
2020	162342.11
2021	575944.08
2022	373319.09
2023	401036.95
2024	318939.45

dtype: float64

Category trends over years

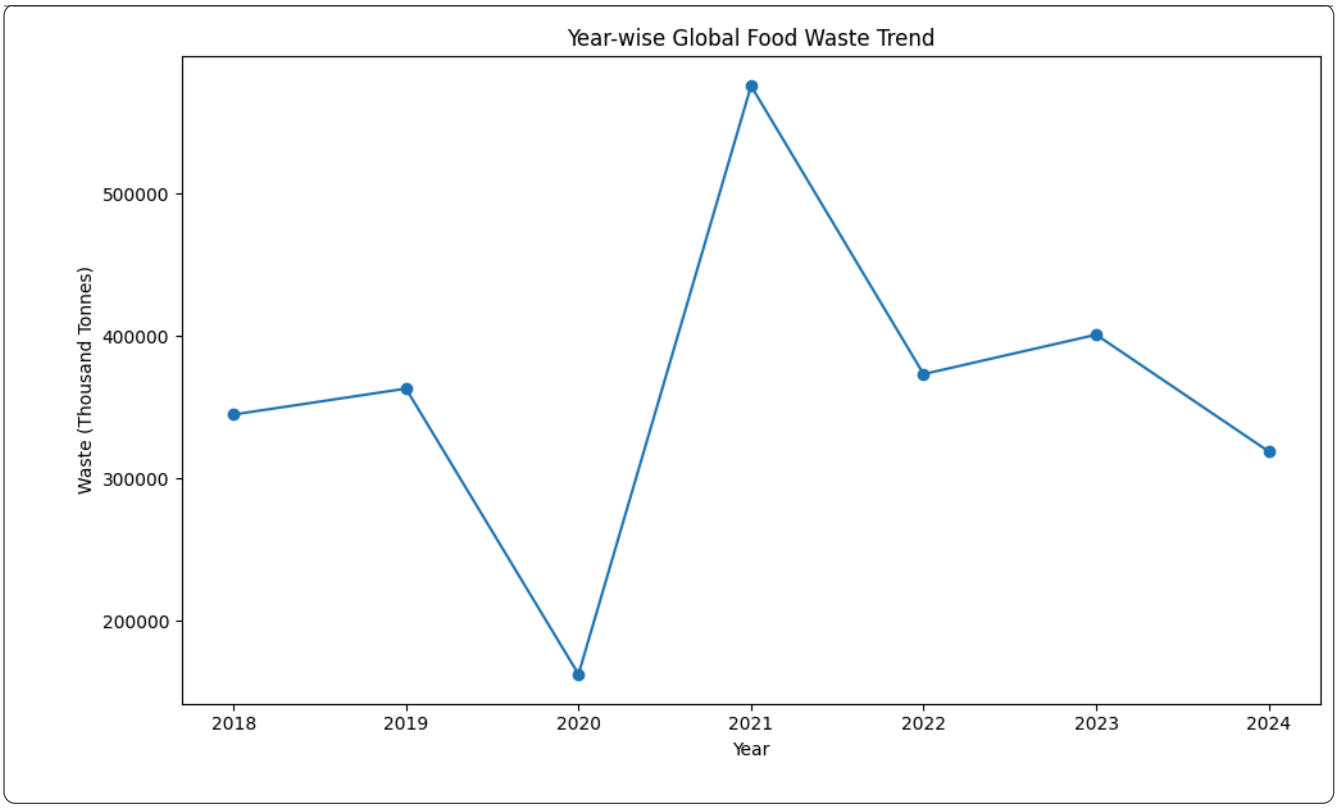
```
category_year = df.groupby(["Year", "Food Category"])["Total Waste (Thousand Tonnes)"].sum()
display(category_year)
```

Year	Food Category	Total Waste (Thousand Tonnes)
2018	Dairy Products	69334.25
	Frozen Foods	66621.49
	Fruits & Vegetables	37952.55
	Meat & Seafood	82736.11
	Prepared Foods	88248.65
2019	Dairy Products	42354.01
	Frozen Foods	66830.17
	Fruits & Vegetables	9189.62
	Grains & Cereals	72045.37
	Meat & Seafood	118542.41
	Prepared Foods	54167.88
2020	Dairy Products	48618.61
	Frozen Foods	11520.27
	Fruits & Vegetables	38008.76
	Grains & Cereals	58100.25
	Meat & Seafood	6094.22
2021	Dairy Products	102094.20
	Frozen Foods	64783.23
	Fruits & Vegetables	79529.80
	Grains & Cereals	198663.24
	Meat & Seafood	67974.76
	Prepared Foods	62898.85
2022	Dairy Products	137364.84
	Frozen Foods	103427.79
	Fruits & Vegetables	10105.91
	Grains & Cereals	29286.15
	Meat & Seafood	46837.13
	Prepared Foods	46297.27
2023	Frozen Foods	108457.79
	Fruits & Vegetables	72488.12
	Grains & Cereals	32675.33
	Meat & Seafood	84134.61
	Prepared Foods	103281.10
2024	Dairy Products	75823.63
	Frozen Foods	70129.71
	Fruits & Vegetables	33240.11
	Grains & Cereals	72568.72
	Meat & Seafood	48834.32
	Prepared Foods	18342.96

dtype: float64

```
yearly_waste = df.groupby("Year")["Total Waste (Thousand Tonnes)"].sum()

plt.figure(figsize=(10, 6))
plt.plot(yearly_waste.index, yearly_waste.values, marker='o')
plt.title("Year-wise Global Food Waste Trend")
plt.xlabel("Year")
plt.ylabel("Waste (Thousand Tonnes)")
plt.xticks(yearly_waste.index)
plt.tight_layout()
plt.show()
```



Waste per million population

```
# Step 1: Waste per country
waste_country = df.groupby("Country")["Total Waste (Thousand Tonnes)"].sum()
# Step 2: Population per country
population_country = df.groupby("Country")["Population (Million)"].mean()
waste_per_million = (waste_country / population_country)
display(waste_per_million.sort_values(ascending=False))
```

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